

Lesson 14

Concentration Inequalities

A concentration inequality
gives us a probability
band on certain random

variables taking atypically
large or atypically small
values (tail probabilities)

$$P(X > a) \quad a \in \mathbb{R}^+$$
$$P(X < -a)$$

1. Markov's Inequality

Assume X is a non-negative r.v.

with $\underline{E[X]} < \infty$, then for any $\alpha > 0$

$$P(X > \alpha) \leq \frac{E[X]}{\alpha}$$

Remark: This inequality is non-trivial when $\alpha > E[X]$.

→ large deviations

$$X > \alpha$$

$B = \{X > \alpha\}$ is an event

$I_B(X)$ is a Bernoulli r.v.

$I_{X > \alpha}(X)$ is a Bernoulli r.v.

$$\forall \alpha \quad I_B(X) + I_{B^c}(X) = 1$$

$$I_{X > \alpha}(X) + I_{X \leq \alpha}(X) = 1 \quad \text{w.p.1.}$$

$$E[X] = E[X(\omega)] = E\left[X I_{X > \alpha}(X) + X I_{X \leq \alpha}(X)\right]$$

$$= E\left[X I_{X > \alpha}(X)\right] + E\left[X I_{X \leq \alpha}(X)\right]$$

$\underbrace{\qquad\qquad\qquad}_{\geq 0}$

$$\Rightarrow E[X] \geq E\left[X I_{X > \alpha}(X)\right]$$

$$X I_{X > \alpha}(X) = \begin{cases} 0 & X \leq \alpha \\ X & X > \alpha \end{cases}$$

$$E[X] \geq E[X I_{X > \alpha}(X)]$$

$$\geq E[\alpha I_{X > \alpha}(X)]$$

$$= \alpha E[I_{X > \alpha}(X)]$$

$$P(X > \alpha)$$

$$P(X > \alpha) \leq \frac{E[X]}{\alpha}$$

As an exercise, prove
Markov ineq. for continuous
r.v.'s using integrals &
def. of expectation.

The image shows two blank, lined pages from a notebook. Each page is enclosed in a brown border and contains ten horizontal lines for writing. The top page has a small notch in the upper right corner. The bottom page is slightly offset to the left and bottom relative to the top page. There are some faint, light-colored smudges or stains on the bottom page, particularly near the bottom left corner.

Question: The "decay rate" is $1/\alpha$ in Markov Inequality.

Can we have a decay rate that is faster than $1/\alpha$?

The answer is YES:

2. Chebyshev's Inequality

Assume that X is a r.v. with expectation μ and variance $\sigma^2 < \infty$, then

$$P(|X - \mu| > k\sigma) \leq 1/k^2$$

$\forall k > 0$

OR

$$P(|X - \mu| > c) \leq \frac{\sigma^2}{c^2}$$

$$k\sigma = c \quad P(|X - \mu| > c) \leq \frac{1}{\left(\frac{c}{\sigma}\right)^2} = \frac{\sigma^2}{c^2}$$

$k = \frac{c}{\sigma}$

Proof:

$$P(|X - \mu| > c) = P(|X - \mu|^2 > c^2)$$

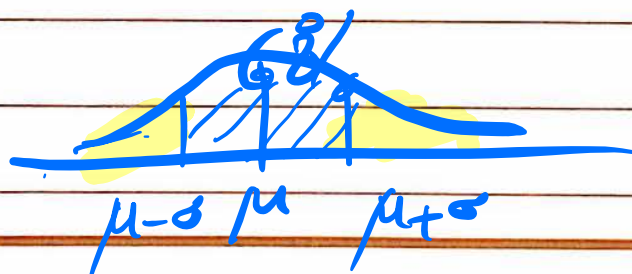
$$\leq \frac{E[|X - \mu|^2]}{c^2} = \frac{\sigma^2}{c^2}$$

By Markov
inequalityPut $c = k\sigma$ to get the other
version

Chebyshev's Inequality
for various k 's

$$k \geq 1 \quad P(|X - \mu| > k\sigma) \leq \frac{1}{k^2}$$

When X is Normal $\approx 32\%$



$$k=2 \quad \mathbb{P}(|X-\mu| > 2\sigma) \leq \frac{1}{2^2}$$

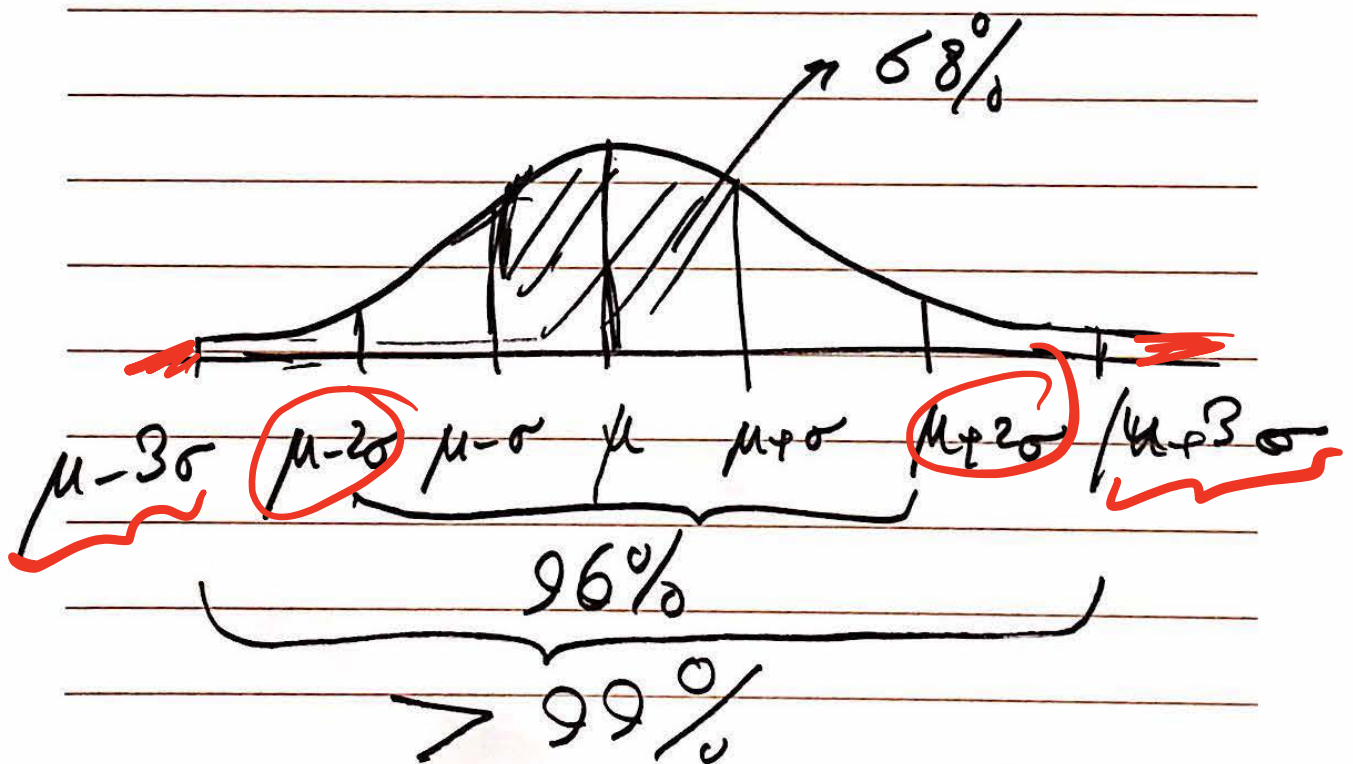
For Normal

$\approx 4\%$ $\swarrow$

$$k=3 \quad \mathbb{P}(|X-\mu| > 3\sigma) \leq \frac{1}{9}$$

For normal

$< 1\%$



Chebyshev Inequality yields a bound that decays as fast as $1/k^2$, an improvement over the basic Markov Inequality.

Question: Can we find bounds that decay faster than ~~$1/\text{polynomials}$~~ , i.e., exponentially?
YES! Chernoff Bound